# Introduction to ARM LPC2148 Microcontroller




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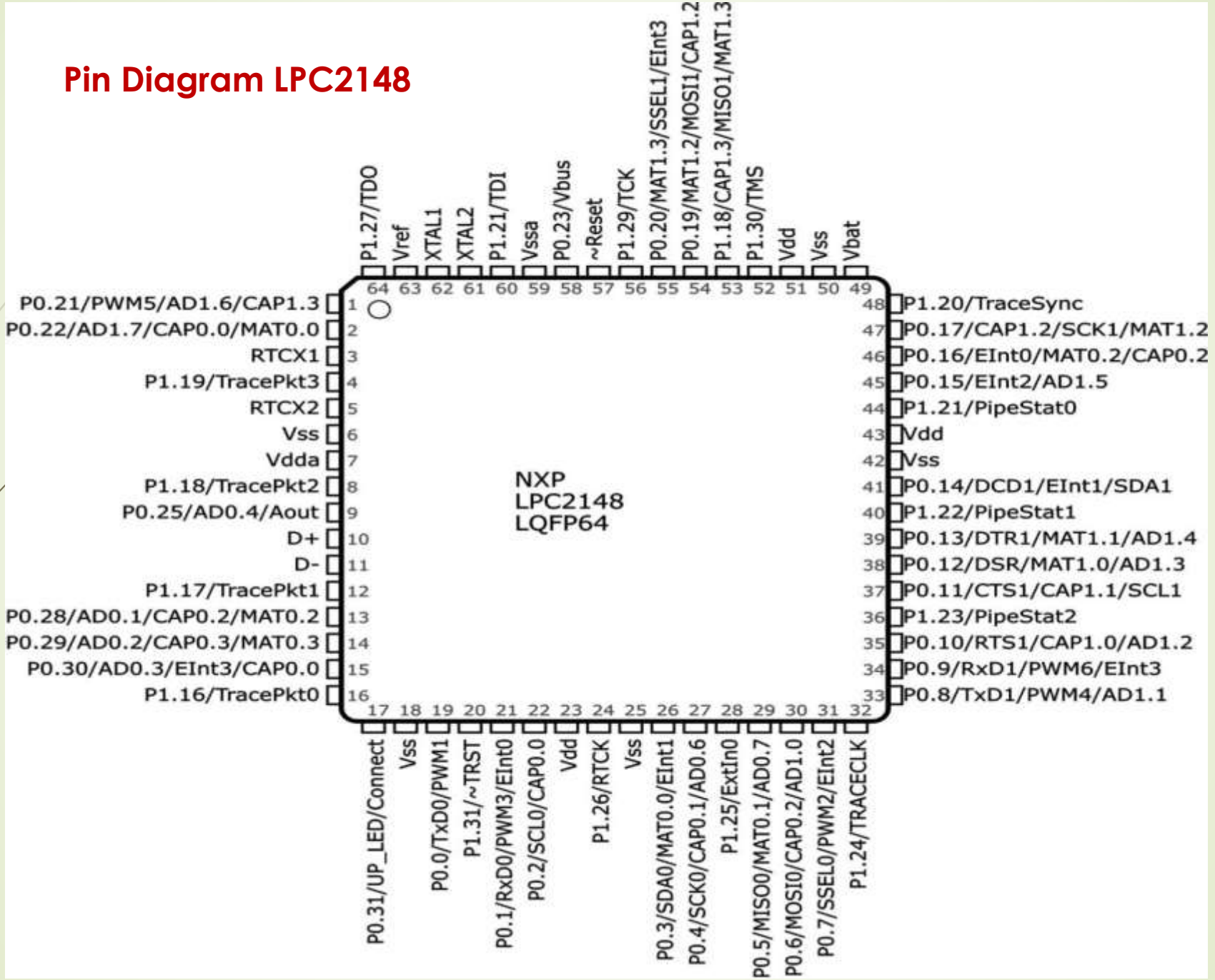
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# Features of LPC2148

- CPU = ARM 7 Core
- Word Length = 32 Bit
- ROM = 512 KB
- RAM = 40KB
- 2 Parallel Port (P0,P1 → 29+16 = 45 GPIO)
- 2 Timers (32 Bit) → 4 Compare, 4 Capture, 6 PWM
- Interrupts sources = 22
- 1 Watch Dog Timer
- 1 RTC
- 2 ADC (AD0,AD1) → 8+6 = 14 Channels (10 Bit)
- 1 DAC (10 Bit)
- 2 -I2C-bus, 2 –UARTs, 2-SPI/ SSP, 1 USB

| Feature | LPC2148 |
| --- | --- |
| Word Length | 32 Bit |
| ROM | 512 KB |
| RAM | 40 KB |
| Parallel Port | P0, P1(45 GPIO) |
| 2 Timers | T0,T1 (32 bit) Capture, Compare, PWM. |
| Interrupts Sources | 22 |
| Watch Dog Timer | 1 |
| Real Time Clock – RTC | 1 |
| ADC | 2 (AD0,AD1) 8+6= 14 channels |
| DAC | 1 |
| UART - Universal Asynchronous Receiver/Transmitter | 2 |
| USB | 1 |
| I2C - Inter-Integrated Circuit | 2 |
| SPI - Serial Peripheral Interface | 2 |

# Pin Diagram LPC2148



# Important Features of LPC2148

- **PACKAGE:**

32-bit ARM7TDMI-S microcontroller in a tiny LQFP64 package with 64 Pins

- **MEMORY:**

40 kB of on-chip static RAM
512 kB of on-chip flash program memory.

- **SPEED:**

Max speed upto 60 MHz operation.

- **ADC**

Two 10-bit A/D converters(AD0 and AD1) provide a total of 14 analog inputs
conversion times as low as 2.44 µs per channel.

- **DAC**

Single 10-bit D/A converter provides variable analog output.

- **TIMERS**

Two 32-bit timers/external event counters
Each timer with four capture and four compare channels
Six PWM outputs
1 Watchdog timer

- **RTC**

Low power real-time clock with independent power and dedicated 32 kHz clock input.

- **Serial Interfaces:**

I2C-bus:

- Two Fast I2C-bus with 400 kbit/s (I2C0,I2C1)

**UART communication:**

- Two UARTs (UART0,UART1)

SPI (Serial Peripheral Interface)and SSP(Synchronous Serial Port)

**GPIO:**

Up to 45 of 5 V tolerant fast general purpose I/O pins in a tiny LQFP64 package

**INTERRUPTS:**

Total of 22 interrupt Sources

Vectored interrupt controller with 16 configurable priorities and vector addresses.

**Speed** :

60 MHz maximum CPU clock speed from programmable on-chip PLL with settling time of 100 µs.

- **OSCILLATOR:**

On-chip integrated oscillator operates with an external crystal in range from 1 MHz to 30 MHz

- **Power saving modes:**

Idle mode

Power-down mode

- **CPU operating voltage :**

Range of 3.0 V to 3.6V (3.3 V ±10 %) with 5 V tolerant I/O pads.

**Other Features:**

- In-System / In-Application Programming (ISP/IAP) via on-chip boot-loader software.
- USB 2.0 Full Speed compliant Device Controller with 2 kB of endpoint RAM.
- In addition, the LPC2148 provide 8 kB of on-chip RAM accessible to USB by DMA.

# Applications of LPC2148

- Industrial control
- Medical systems
- Access control
- Point-of-sale
- Communication gateway
- Embedded soft modem
- General purpose applications